

Raghav Chalapathy

EDUCATION	University of Sydney, NSW , Australia <i>Doctorate of Philosophy (PhD) , School of Computer Science Jul' 15 - December' 19</i> Illinois Institute of Technology, Illinois , USA <i>Master of Information Science and Security , School of Computer Science Feb' 11 - Jan' 13</i> R.V. College of Engineering, Bangalore, India <i>Bachelor's of Telecommunication Engineering, Dept. of Telecommunication Aug' 00 - Dec' 04</i>
INDUSTRY CERTIFICATIONS	International Information System Security Certification Consortium <i>Certified Information Systems Security Professional CISSP, Jul' 13 - December' 16 (License: 435950)</i> Cloudera Data Scientist Certification <i>Certified Data Scientist Since Jul' 18</i>
EXPERTISE/ RESEARCH	Deep Learning for Anomaly Detection, Natural Language Processing, Time series forecasting.
RECENT PUBLICATIONS:	Linsey Pang, Wei Liu, Lingfei Wu, Kexin Xie, Stephen Guo Raghav Chalapathy, "Applied Machine Learning Methods for Time Series Forecasting" <i>31st ACM ACM International Conference on Information & Knowledge Management October (CIKM-2022)</i>. Frances Cameron-Muller, Weeraddana, Dilusha, Raghavendra Chalapathy, Nguyen Lu Dang Khoa. "Dual-Stage Bayesian Sequence to Sequence Embeddings for Energy Demand Forecasting" <i>25th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD-2021), May 11-14, 2021</i>. Raghavendra Chalapathy, Nguyen Lu Dang Khoa, Sanjay Chawla. "Robust Deep Anomaly Detection" <i>Hands On Workshop : KDD '20: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining August 2020 Pages 3507:3508</i> https://doi.org/10.1145/3394486.3406704. Raghavendra Chalapathy and Sanjay Chawla. "Deep Learning for Anomaly Detection: A survey" <i>arXiv preprint arXiv:1901.03407 Pages 1:50 (2019)</i>. For More publications and citations kindly refer my: Google scholar profile.
AWARDS & ACHIEVEMENTS	Part of Applied AI team at Walmart GlobalTech winning both Edelman and Inform's Award April '23 Awarded Bronze Medal in Academic Track Contributions @ Global AI Challenge - Feb'22 Awarded the Intelligent Transport System [ITS] Australia National Award - Dec '20
TECHNICAL SKILLS	Software and Programming Languages: Python (scikit-learn, numpy, scipy, pandas, NLTK, gensim, keras, pytorch, tensorflow), R, SQL, Hadoop (Pyspark, MapReduce), Linux, Microsoft Excel, Javascript, C, C++, Java, L^AT_EX. Machine Learning: classification, regression, clustering, feature engineering, deep learning models for anomaly detection. Statistical Methods: time series, regression models, hypothesis testing and confidence intervals, principal component analysis and dimensionality reduction. Selected Coursework: Natural Language Processing, Machine Learning and Data Mining, Linear Algebra, Probability and Statistics.
CURRENT EXPERIENCE	Organisation : Walmart Global Tech Designation : Staff Data Scientist at Walmart Labs Feb '22 - till date

Responsibilities: Lead team of data scientist's, machine learning engineers by interfacing business requirements obtained from internal stakeholders, conduct proof of concept research and participate in publishing novel methods adopted in external conferences and enable brand building.

Project: Building Next Generation Geo-Demand Forecasting Platform Aims to build platform which enables demand signals for downstream operations and strategic decision teams to build models for optimal allocation of resources and help reduce infrastructure, maintenance costs

PREVIOUS PROFESSIONAL & EXPERIENCE

Organisation : CSIRO's Data61

*Designation : Research Fellow at Transport and Analytics Group, Data City Team, The Commonwealth Scientific and Industrial Research Organisation (CSIRO), Eveleigh, New South Wales, Australia
Apr-2019 - Feb '22*

Projects, Roles & Responsibilities:

- Involved in developing anomaly detection models for structural health monitoring and incident detection within transport network data. Client: Sydney Harbour Bridge, NSW Transport.
- Develop models for predicting Heating, ventilation, and air conditioning (HVAC) load and solar energy generation within regional hospital.
- Supervising PhD student: Identifying concept drift and detecting anomalies within data streams.

Organisation : NSW Ministry of Health

Designation : Data Scientist | Research Officer: NSW Ministry of Health. July '15 - March '19

Projects Responsibilities & Quantifiable results:

- Finding service usage patterns and investigating the prevalence of low-acuity care provided by NSW-Ambulance and Emergency department by applying data mining and machine learning and natural language processing algorithms.
- Investigation revealed about 17% of transported patients reaching emergency department hospitals presented with low risk back-pain as chief complaint, which suggest the need to explore, safer cost effective models of care in the community.

Organisation : Huawei Technologies-Bangalore India

Designation : Senior Technical Leader

August '06 - Jun '15

Projects Responsibilities & Quantifiable results:

- Involved in building models identifying software failures using machine learning classification techniques and conduct software security audits. Lead a design and development team (group of 5) to develop and automate tools for general software and security testing.

TEACHING PROJECT MANAGEMENT

Tutoring Master of Data Science Courses

Role : Tutor , University Of Sydney, Post Graduate level Courses: [Master of Data Science](#) Fall '16 - Fall'19

Capstone Projects-COMP5703

Role : Project Manager, Examiner University Of Sydney, Graduate Course: [Capstone Projects](#) Aug '18 - Fall'19

COMMUNITY SERVICE

Member: of Artificial Intelligence and citizen science group @ [Australian Citizen Science Association](#).

Member: Rational AI University A non-profit arm of [rationalai.org](#) for teaching AI and machine learning skills to support low income group individuals and less fortunate kids in India.